



PRODUCT NAME

Industrial-grade Neck Pain Cream Formula

Description

This expertly formulated neck pain cream provides targeted pain relief with anti-inflammatory properties, all while being fast-absorbing and non-greasy. Designed for industrial use, it is crafted to soothe muscle discomfort effectively.

Technical Specifications

pH Level:	7
Viscosity:	High, typical of a rich cream
Shelf Life:	24 months
Batch Size:	500L to 1000L
Processing Time:	Approximately 2 hours
Temperature:	75°C during heating phases
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Stainless steel mixing tank, high shear mixer, heating system, pH meter
Certification:	ISO 9001, GMP compliant

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Caprylic/Capric Triglyceride	Caprylic/Capric Triglyceride	10.0%	Emollient
3	Glyceryl Stearate	Glyceryl Stearate	6.0%	Emulsifier
4	Cetyl Alcohol	Cetyl Alcohol	5.0%	Thickener/Emulsifier
5	Menthol	Menthol	5.0%	Cooling Agent/Pain Relief
6	Methyl Salicylate	Methyl Salicylate	4.0%	Anti-inflammatory/Pain Relief
7	Camphor	Camphor	3.0%	Pain Relief
8	Carbomer	Carbomer	2.0%	Thickener
9	Triethanolamine	Triethanolamine	1.8%	pH Adjuster
10	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
11	Methylparaben	Methylparaben	1.0%	Preservative
12	Aloe Vera Extract	Aloe Barbadensis Leaf Extract	1.2%	Soothing/Anti-inflammatory
Total			100%	

Manufacturing Process

Phase 1: Phase A

1. Combine Water and Carbomer. Mix well until carbomer is fully hydrated.
2. Heat the mixture to 75°C.

Phase 2: Phase B

1. In a separate container, combine Caprylic/Capric Triglyceride, Glyceryl Stearate, Cetyl Alcohol. Heat to 75°C until uniform.
2. Add Menthol, Methyl Salicylate, and Camphor to the oil phase until dissolved.

Phase 3: Phase C

1. Slowly add Phase B into Phase A with continuous mixing.
2. Cool the mixture to 40°C while stirring.

Phase 4: Phase D

1. Add Aloe Vera Extract, Phenoxyethanol, and Methylparaben into the cooled mixture.
2. Adjust the pH using Triethanolamine to reach exactly pH 7.
3. Mix until homogenous.

Required Equipment

Stainless steel mixing tank, high shear mixer, heating system, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001, GMP compliant

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 500L to 1000L.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of cream to the neck area and massage gently until fully absorbed. Use up to three times daily as needed for pain relief.